

Automatically identifying customer needs in user-generated content using token classification

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ABSTRACT

Users generate tremendous amounts of data on the Internet every day. This so-called user-generated content (UGC) is valuable input for organizations since it may include individual experiences, opinions, and desires with respect to the products and services they offer. To automatically process UGC, automated techniques, typically referred to as Needmining, have been developed. Existing Needmining approaches extract customer needs from UGC by binarily classifying unstructured textual data into need-content and no-need content. However, they are not able to extract the specific needs. We address this research gap by developing a decision support artifact that re-conceptualizes Needmining from a binary classification problem to a token-classification problem to extract specific needs from informative content. To achieve this, we break down customer needs into components, i.e. attributes and characteristics and develop a token classification artifact. The artifact accurately identifies the need-components and, therefore, can identify specific customer needs in user-generated content. We organize and discuss the value of the artifact's output and further enrich the model with sentiment data to distinguish relevant needs. If applied, the artifact can realize efficiency gains for decisionmakers in the field of product development as it automatically and quickly identifies relevant consumer needs.

1. Introduction

Web 2.0 sites and social media platforms are generating a tremendous amount of data every day. This so-called digital user-generated content (UGC) reflects individual experiences of users that they willingly share through various channels on the Internet. Examples include posts on Twitter, videos on YouTube, pictures on Instagram, but also film critiques, food recipes, and product reviews on e-commerce platforms such as Amazon and Alibaba.

Many organizations have realized that UGC contains valuable insights into customers and their needs and, hence, should be systematically analyzed and explored. For example, UGC can support decision makers with analyzing products' competitive advantage [1] and predicting firm equity value [2]. However, UGC is mostly considered as informal input [3], likely because the manual analysis of UGC is both cost- and time-intensive due to its high volume and unstructured nature. There have been several attempts to provide automated support in this regard. In literature, this task of automatically eliciting and gathering customer needs from UGC has been established as 'Needmining' [4]. The

idea is that information extracted through Needmining can complement insights that are generated through traditional business practices such as surveys, expert panels, and field observations [5] and, in this way, support decision making in the context of product development and innovation.

Existing Needmining approaches conceptualize Needmining as a binary classification problem, i.e., they aim to determine whether a given UGC contains a need or not. One of the first studies was conducted by Kuehl et al. (2016) [4], who investigated the capability of various machine learning algorithms to separate need-content from no-need-content in micro-blog data from Twitter. Later, several other authors followed up and investigated to what extent machine and deep learning can be used to separate informative content from uninformative content in UGC [5–7]. What all existing Needmining approaches have in common is that they are not able to extract specific customer needs. By conceptualizing Needmining as a binary classification problem, they are only able to detect whether UGC contains a need. Kühl et al. (2020) [8] extended this approach with a multi-label classifier to sort the sentences into predetermined categories. However, this approach only identifies

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needs that are present a priori in the training sample. This severely limits its usefulness for decision makers, since unknown needs are most likely to offer a competitive advantage [5].

This study follows design science research (DSR) [9] to propose a novel Needmining approach that overcomes this limitation of existing works. To achieve this, we reconceptualize the Needmining problem from a binary classification task to a token classification task. This allows us to take the problem from the macro-level (classify reviews and sentences) to the micro-level (classify words and word sequences) and extract the exact words that describe a given customer need. Our approach provides decision makers with the ability to evaluate detected needs through their corresponding textual content, employing techniques like sentiment analysis or estimating the importance by counting their occurrence in the data. By extracting specific needs, we differentiate ourselves from studies that classify sentences based on pre-determined categories, like Kühl et al. (2020) [8], and provide the ability to uncover previously unknown needs.

2. Problem statement

Needmining makes extracting customer needs from UGC viable on a large scale. However, there is still much potential to increase the efficiency at which this is achieved. The common denominator between all previous studies is the amount of manual (human) labor that is required to identify the exact customer needs customers express within their UGC. There is also a research gap that appears in the literature on more than one occasion that calls for further automation in the analysis steps following the binary classification of UGC [4,5]. Our study aims at addressing this research gap by increasing the efficiency of Needmining by further automating the process by manually extracting specific needs and meaningfully organizing them. This provides decision makers with an instant overview of customer needs across large bodies of UGC.

Large bodies of UGC come with specific technical challenges that need to be addressed. As stated above, this study moves the Needmining problem from binary classification to token classification. However, we can still look towards previous methods for advice when dealing with this type of data. Previous binary Needmining Methods adapt various machine learning models to analyze unstructured data. Additionally, recent developments in machine learning, specifically the introduction of transformers [10] and transfer learning [11] significantly boosted the performance of natural language processing giving language models a significant advantage when dealing with textual data [12]. Like other machine learning models, once trained to perform a specific task, language models are also easily scalable to large datasets with little additional computational effort. Therefore, we chose to implement a transformer-based language model to reflect state-of-the-art performance and compete with other methods.

Based on the technical challenges of large bodies of UGC, we suggest training our token classifier to detect product attributes and characteristics based on pre-trained word embeddings. This approach fits the problem specified within this research the best because it allows expert knowledge to be integrated into the discovery of new needs through training a model based on expert input.

3. Related literature

3.1. Customer needs

Customer needs are an important source of information for managers in marketing, innovation, and product development. They inform managerial decisions on product design, which can affect the success of individual products and, by extension, entire businesses. Although there are many definitions for needs, we focus on those derived from marketing and innovation. In marketing, customers play a vital role, resulting in definitions centered around the customer. One common view considers customer needs to be a detailed, self-expressed

description of the benefit the customer expects a product or service to fulfill without being only a solution or physical measurement [5,13]. Another common view separates needs into three distinct categories, i. e., needs, wants, and demands [14–17], where needs are basic human requirements, wants are the specific objects towards which needs draw the customer, and demands are both the ability and willingness of individuals to pay for an object [16].

In the context of Needmining, there is no benefit in splitting customer needs into needs, wants, and demands. Customers create user-generated content specifically to express themselves using their own words. In some contexts, such as product reviews as an explicit form of UGC, the definition marketers use for needs fits even more closely. Customers use product reviews to describe their experiences with any given product. They usually include information regarding their expectations from a product, if the product fulfilled their expectations, which needs they expected the product to fulfill, which needs the product did not meet, and sometimes what new needs arose from experiencing the product.

To fully understand customer needs, it is also relevant to understand some of their applications. There is a notion of products as a collection of attributes which are designed to address customer needs by approaching specific target values for each attribute [18]. This is a key principle of conjoint analysis, which is an analytical method to determine the optimal product design by finding the perfect combination of attributes and attribute target values (or attribute dimensions). One conjoint analysis variant specifically identifies product attributes which lead to unique benefits experienced by customers [19], enhancing the link between customer needs and product attributes.

These applications of needs are strongly compatible with Needmining theory. Attributes and attribute dimensions are frequently featured in UGC when customers express their experiences with products and criticize their shortcomings by addressing lacking features. For the remainder of this paper, we refer to attribute dimensions as characteristics given that they describe the characteristics of a product attribute.

3.2. Needmining

Using customer needs for innovation is increasingly becoming an important topic in information systems literature [1,20,21]. Kuehl et al. (2016) [4] originally coined the term Needmining to describe their artifact which extracts need containing content from UGC. A range of studies followed this approach using various methods. However, the underlying mechanic in Needmining-related studies is always a binary classifier that removes no-need-content from need-content. In more general words, they separate informative content from uninformative content [5].

There are a range of classifiers that have been used for the purpose of Needmining, such as support vector machines [4,7], naive Bayes classifiers [4], tree-based classifiers [4], and neural networks [5,6,22]. Empirical evidence suggests that the application of such Needmining tools can support product developers and innovation managers in their decision making by identifying novel needs that otherwise prove to be “a needle in a haystack” [5] in the vastness of user-generated content. Timoshenko and Hauser (2019) [5] further support this by both putting an economical value to Needmining and identifying, in their own words, “hidden opportunities”, which are customer needs that the analysis of customer reviews reveals, but that expert product developers do not find. Additionally, Kühl et al. (2019) [23] compare needs from the literature to needs expressed in Twitter micro blog data. They find that the frequency distribution of needs differs significantly between these sources. Yet, they highlight the relevance of tracking social media for customer needs to enable firms to respond quicker to market changes.

One of the key research gaps identified by studies that classify user-generated content is that manual analysis is required to extract specific needs [4,5]. However, so far, we have only discussed Needmining as a binary classification problem. This means that available automated

methods can only separate informative content from uninformative content, while further processing and analysis of customer needs remained a manual effort. We argue that the scope of Needmining needs to include automation of downstream steps as well. Specifically, we suggest adding a second pillar to Needmining that deals with the automated extraction of specific customer needs from UGC. In the following, we discuss studies that deal with this second pillar of Needmining and relate to our study.

Kühl et al. (2020) [8] contribute to the need extraction process by creating an artifact that assigns UGC to predetermined need categories using supervised machine learning. This is a crucial step towards a fully automated Needmining approach including the extraction of specific needs. However, experts must still identify relevant categories for every research domain, which is a significant amount of manual labor. Other studies apply the previously discussed view of products as a vector of attributes, for example, by identifying and evaluating attributes using part-of-speech tagging and dictionary-based sentiment analysis [24]. They address key problems with mining user-generated content, notably the large amount of available content, the lack of relevant information in the available content, and, by extension, the lack of oversight that both customers and producers may experience when analyzing such content. Roelen-Blasberg et al. (2022) [25] also use part-of-speech tagging combined with dependency parsing to extract noun-adjective combinations from UGC. Subsequently, they group them together with word embeddings, and evaluate the importance of the attributes with sentiment analysis and conjoint analysis. This approach functions on the word level; however, we assume that in the context of word-level need mining, this would lead to many false-positive identifications, as there are no filters to determine which of these combinations represent relevant needs. Comparably, Lee and Bradlow (2011) [26] identify product attributes and characteristics from product reviews represented as a list of positive and negative aspects using a combination of methods such as transforming sentences to word vectors, cosine-similarity, and K-means clustering. Their findings support the notion that traditional methods of market research, such as expert panels and surveys, may well overlook important product attributes that could otherwise inform managerial decisions. Finally, Wang et al., (2022) [27] apply a clustering approach to an embedding space to identify the relationship between “engineered attributes”, which product experts manually extract after automatically pre-processing textual data, and more relevant product benefits.

Other indirect Needmining studies include approaches that use network analysis and sentiment analysis to identify and associate relevant keywords with particular aspects of customer service [28] and finding need-solution pairs using what they call semantic network analysis and word-space models (i.e. word embeddings) to filter UGC [29]. While these studies are also relevant in the Needmining context, they require more manual labor to identify needs compared to the other more automated methods. However, both provide valuable insights into post-processing of already identified customer needs, specifically sentiment analysis is useful to quantify already identified needs for further steps in product development.

Finally, it could be argued that topic modeling methods such as LDA [30] could function for word level Needmining because the words this approach uses to describe the topics could represent needs. However, the effectiveness of this type of information has its drawbacks because significant labor would be required to analyze the latent topics created by such a model and to turn the information into needs.

3.3. Token classification

In natural language processing, tokenization refers to the segmentation of text into smaller basic units of processing called tokens [31]. Often tokens represent word-level units, while recent deep learning research uses sub-word tokens [11,32]. The tokenization of the text is a prerequisite for its further processing in many tasks, for example, for syntactic analysis by parsers, in text mining, or in information retrieval.

Token classification is a method to classify a given span within a string.

In the string “The weather is nice”, each word gets a positional encoding. A span is a sequence of these word positions. If ‘The’ gets position one and ‘nice’ gets position four, then the span 1:2 would represent the words ‘The weather’. In contrast to text classification, the model predicts every token in the text instead of a single prediction for the whole text (see Fig. 1). The best-known application of this method is ‘Named Entity Recognition’. Named Entity Recognition is the process of identifying specific substrings within a string and classifying them according to what the model was trained on, for example, persons, locations, numeric values, firms, or countries [33]. Assuming a model was trained to detect these categories, we would expect it to identify three spans, or entities, in the sentence “Michael Jeffrey Jordan was born in Brooklyn, New York”: Michael Jeffrey Jordan (person), Brooklyn (location), and New York (location) [34].

There are many different architectures that can be used for named entity recognition, however, deep learning and neural networks seem to set the highest benchmarks in Named Entity Recognition tasks [35]. In theory, there is no limit to what an entity could be if there is concise training data. Previously, studies used Named Entity Recognition for part-of-speech tagging [36], and for extracting biomedical entities from domain-specific literature [37,38]. However, Named Entity Recognition plays a major role in a wide range of applications such as information extraction, question-answering, machine translation, text summarization, text clustering, opinion mining, and more [39]. Because Needmining is an application of information extraction, we reconceptualize the Needmining problem from binary classification to a token classification problem. In the following section, we introduce our Needmining approach on a conceptual level.

4. Research methodology and artifact design

4.1. Design science research

This design science project is guided by the design science research process model of Kuechler and Vaishnavi (2008) [40]. Fig. 2 illustrates our application of this process model and shows the structure we use to produce research steps towards a relevant artifact that contributes to the knowledge base. The first and arguably most crucial step is to develop a clear understanding of the problem by gathering insights from stakeholders and conducting an in-depth literature analysis. A clear understanding of the problem grounded in the Needmining domain, the composition of customer needs, and the application of customer needs are key to an impactful contribution to the knowledge base. Next, we develop initial solutions to the problem. This process step is informed by the understanding of the problem we developed in the previous step and our extensive literature review. We translated these solutions into artifact features, which we designed to address the existing problems with our solutions directly. Finally, we evaluate and discuss the artifact based on suitable performance metrics and summarize the results of our research.

4.2. Requirements

From a practical perspective, conversations with industry experts from firms specializing in data-driven innovation revealed several problems with extracting relevant information from product reviews. First, they reaffirmed that the massive amount of content is an issue for product developers because the input required for an analysis is too costly. This encompasses not only reading reviews, but also manually analyzing, extracting, and organizing the information within the content. The experts mentioned that it is possible to get insights into drivers of success or failure of certain existing products by reading only a small portion of the reviews with the best and with the worst ratings. However, research shows that this could mean missing out on ‘hidden needs’ relevant for new product development because they go lost in the

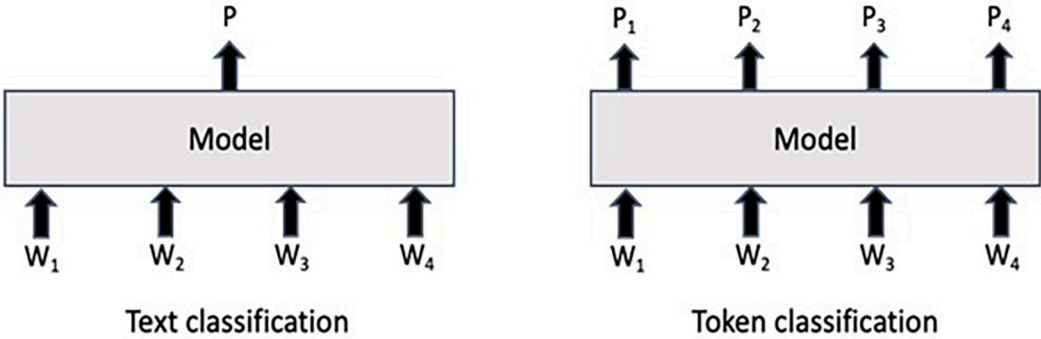


Fig. 1. Conceptual Model Architecture of Text Classification and Token Classification.

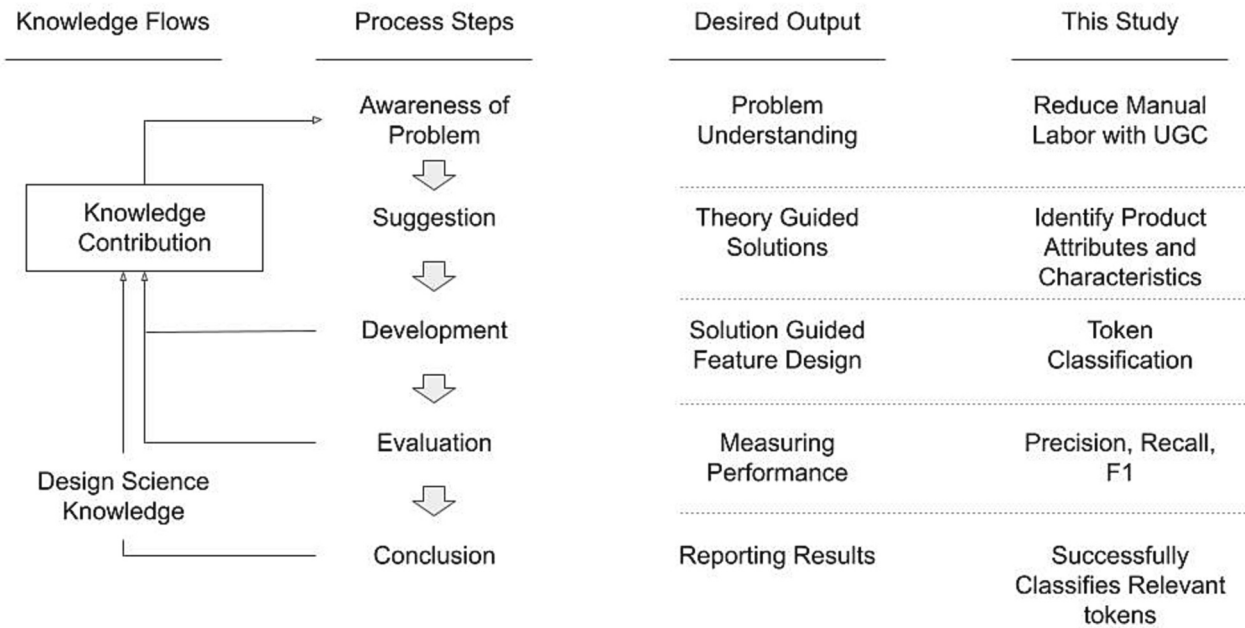


Fig. 2. Design Science Research Process Based on Kuechler and Vaishnavi (2008).

broader mass of unread reviews [5].

Next, we must understand what managers seek in UGC that they can use to make informed decisions. In previous Needmining studies, this is the process step where manual labor extracts the customer needs from the UGC that the algorithm identified as needs containing. Thus, we need to understand what constitutes a specific customer need. To achieve this, we combine our problem understanding with the literature on customer needs to find out how relevant customer needs manifest themselves in UGC.

We combine the view of products as a vector of attributes and characteristics with the definition of customer needs as the benefit a customer expects to receive from a product. In the context of product reviews, we find that customers express their needs by specifying desired attributes and characteristics. Attributes typically describe a feature of a product, while characteristics describe how the feature should be implemented. As an example, consider the following sentence from a review of Apple AirPods of the third generation: “It is a shame that the case can still not be charged wirelessly”. In this sentence, the customer addresses the ‘case’ as the attribute and ‘charged wirelessly’ as the characteristic to express that they in sum have a need for a case with wireless charging capabilities. While the sentence contains other information, such as sentiment (‘shame’) and negation (‘can still not’), the characteristic-attribute combination captures the essence of the need while the additional information only provides context.

It is important to note that individually neither the attribute nor the characteristics are customer needs. However, because we can disentangle the need into two parts it is possible to organize the information in such a way that we can quickly identify everything customers say about a certain attribute. In the example of the AirPods case, we can also capture information on other needs, such as their *weight*, *color*, *battery life*, and more. This illustrates how we can break down customer needs into sub-parts to identify useful information for product developers.

Knowing the components of customer needs allows us to specify a machine learning approach to extract relevant attribute-characteristic combinations, which moves beyond the binary approach of previous Needmining methods. To do so, it is necessary to identify approaches that function at the correct granularity. In the literature review, we already identified token classification as a method designed for this exact purpose, as it allows us to train machine-learning models on custom data to classify specific words within sentences. This is particularly effective because of new insights on advancements in natural language processing thanks to transformers and transfer learning. To summarize, the artifact must be capable of extracting distinct customer needs from large bodies of UGC and then meaningfully summarize these needs to provide an easily accessible overview.

4.3. Model specification

In this chapter, we develop the necessary steps for the artifact to achieve its main objective, which is identifying exact customer needs from UGC. We summarize these steps in Fig. 3 by providing the generalized steps and combining them with a specific example. To begin, the approach requires input, which goes through a process to produce an output that aids decision makers in the product development and innovation process. Thus, the first step requires some form of *content sampling* as an input.

Product managers can sample UGC from a range of platforms, such as textual content like reviews from online shops (e.g. Amazon, Walmart, Costco), blog posts (e.g. Twitter, Reddit), transcribed video reviews and comments to these videos (e.g. YouTube). With sources like this, it is possible for decision makers to create highly diverse data or specific data, for example by considering reviews for a single product category across a range of competitors, or by considering only the product(s) of a main competitor, respectively. To get to the desired output, i.e. organized specific customer needs, the data must undergo a transformational step that extracts relevant characteristic-attribute combinations. As previously discussed, this is the step where we reconceptualize Need-mining from a binary problem to a token-classification problem. Because we employ a machine-learning algorithm, the first step of the process is for the artifact to convert the unstructured textual data, i.e., the product reviews, into a machine-interpretable format for the downstream task. To achieve this, we first apply *sentence tokenization* to each review to split the reviews into individual sentences effectively. For this, we use the Natural Language Toolkit.¹ Later, this step allows the decision makers to quickly identify the origin of a need for cases where additional information may be required. We also tried tokenizing based on whole reviews instead of sentences but found that it degrades the performance of the model. Furthermore, operating on the sentence level has the advantage of a more granular analysis for subsequent steps such as sentiment analysis because we maintain more information.

The next step is '*token embedding*'. Here, we transform the review sentences into tokens. For our embedding strategy, we used Byte-Pair Encoding (BPE). BPE split words further into sub-words to generate the final tokens. For example, it can split the word "annoyingly" into the token *annoying* and *ly*. We chose this tokenization strategy since it is able to keep the meaning of the composite word construct. Out of those tokens the final embeddings are created, which are a contextual representation of words in a real-valued vector space where words with a similar meaning are located more closely to one another [41]. To transform the document, we put the vectorized documents through a pre-trained transformer. We tested various popular pre-trained transformer-based models for this step (BERT [11], RoBERTa [32], XLNet [42], ALBERT [43], DistilBERT [44]). Pre-trained embeddings [11] are the key to modern natural language processing models being both computationally more efficient and better performing.

Next, the approach moves on to '*token classification*', where a neural network makes the final predictions. First, the sub-word tokens are pooled using a mean pooling layer to generate word embeddings. These embeddings are then feed into a three layer fully connected neural network which outputs the final prediction. The token classifier needs to identify product attributes and attribute characteristics. However, it must do so only in cases where customers explicitly discuss these in a review setting and not in generic sentences. It is of no use to product developers when customers make generic comments about attributes without assigning quantifiable value to them. The plethora of edge cases that can occur make it obvious why rule-based systems are not suitable for this type of task. For example, consider the following part of a review about a solid-state hard drive as found on Amazon: "Fast boot up time, averaging 18 seconds from a cold boot". In this sentence, "boot" may be

considered an attribute while "fast" is the attribute characteristic. However, in the second half of the sentence, we again come across the attribute "boot" and the potential characteristic "cold" which provide a setting in which we do not want the classifier to detect these tokens.

This is where the combination of transformer based contextual token embeddings and a neural net become relevant, as the positional encoding of the individual tokens relative to each other teaches the neural network in which context tokens may or may not be classified. Effectively, the artifact should classify only relevant spans as either attribute or characteristic. To achieve this, we require a high precision in the annotated training data to reflect only relevant tokens.

The final step of the artifact is '*need extraction*'. In this final step, we organize the predicted attributes and characteristics to provide a quick overview of all customer needs. The attributes are the features of a product that consumers want to discuss in their reviews. For decision makers, they are useful information because they hint at what the customers want to change about the product. Thus, we create a frequency overview of mentions for each attribute, to identify what customers talk most about. Next, we combine each attribute with their respective characteristics, i.e. all characteristics that appear in a sentence where the attribute appears. This generates an overview of characteristics associated with every attribute. For additional information, we add a list of every full sentence where each identified attribute-characteristic combination appears. This provides a quick overview for decision makers to reiterate in what context consumers discuss the identified needs and differs strongly from existing token classification pipelines.

To summarize, the artifact provides several conceptual and technical contributions. First, we add to the Needmining literature by extracting specific customer needs by splitting this abstract concept into more explicit components (i.e., attributes and characteristics), and identifying them using token classification. Vice versa, the technical contribution to the token-classification literature is that we test the capabilities of state of the art, language models powered token classification for abstract concepts such as customer needs. To achieve this, the artifact consists of the five steps displayed in Fig. 3, i.e., *content sampling*, *sentence tokenization*, *token embedding*, *token classification*, and *need extraction*. The input is a carefully chosen sample of UGC, in this case customer reviews. The transformative steps of sentence tokenization and token embedding are necessary before performing token classification, the most important step of the artifact. It automatically identifies attributes and characteristics, which combine into full customer needs. In the final step, which contributes mostly to the usability of the classified content, we meaningfully organize the needs for decision makers to have a quick overview of all the relevant information, essentially providing a decision support system for managers.

4.4. Training

Training a token classifier requires a dataset of appropriately annotated text. Guided by our problem understanding, we identify customer reviews as the most relevant source of consumer needs for product developers. Amazon, as one of the globally leading e-commerce platforms, is an excellent source for massive amounts of consumer reviews stretching across a multitude of product categories. In agreement with domain experts, we selected the second generation of "Apple AirPods" as a product to train and test the artifact. As of today, second generation Apple AirPods accumulated over 400,000 product reviews on Amazon. During November 2019, we scraped 1036 of the most recent product reviews in accordance with Amazon's terms of usage. We paid a group of student workers to annotate the reviews. The students were briefed on the problem understating based on the literature. This ensures that they fully understand both the overall problem of this paper and how we understand customer needs into attributes and characteristics. With this knowledge at hand, each tagger received a set of full customer reviews.

We analyzed 1036 reviews comprising 3059 sentences, with Fig. 4 showing the distribution of attributes and characteristics. There are

¹ <https://www.nltk.org/>

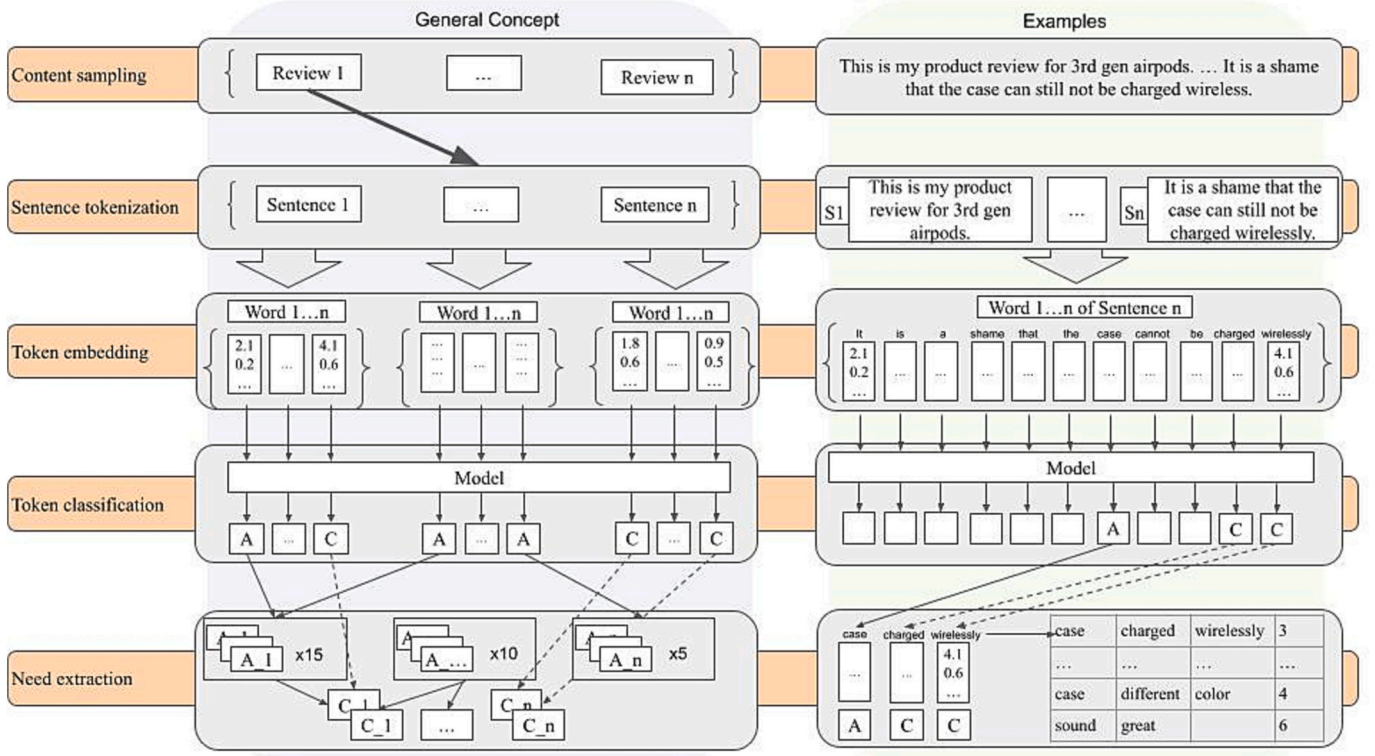


Fig. 3. Artifact overview.

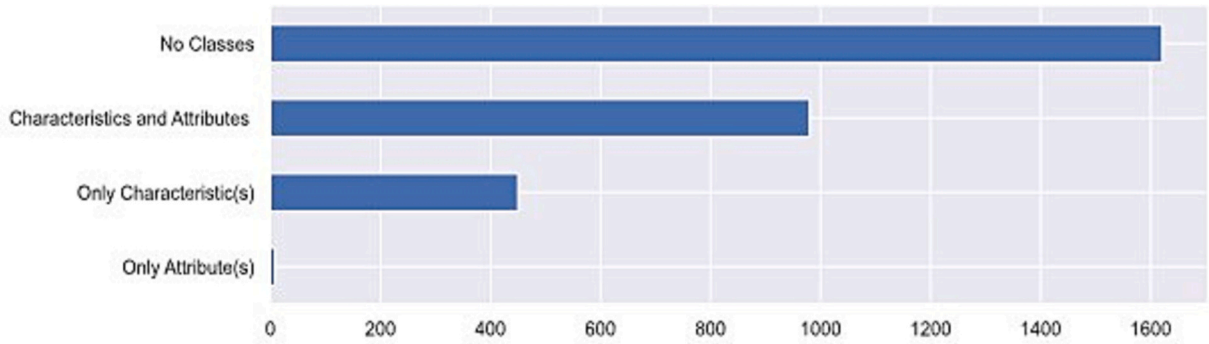


Fig. 4. Token Distribution in Sentences.

1438 sentences containing at least one annotation, while 1621 have no annotations. Although the need and no-need sentences are balanced, there are overall more characteristics than attributes in the data. Some sentences identify relevant characteristics but lack attached attributes, likely referring to the product in general or containing only descriptive words, such as “They stay in the ear well” or “lightweight & comfortable.”

Table 1 provides an overview of how identified tokens are composed within the classes, i.e., how many words are in single spans, and how the classes are distributed to sentences, i.e., how many attributes and

Table 1
Descriptive statistics for Sentence composition.

Class	Level	Q25	Q50	Q75	Max	mean
Attributes	Token per Span	1	1	2	9	1.51
	Spans in Sentences	0	0	1	4	0.41
Characteristics	Token per Span	1	2	2	12	2.03
	Spans in Sentences	0	0	1	5	0.59

characteristics appear within sentences. On average, a token annotated as an attribute contains fewer words than those annotated as characteristics. Moreover, sentences in general appear to contain a larger number of characteristics than attributes.

5. Model evaluation

The evaluation of our approach consists of several parts. First, we present the performance based on commonly used metrics to quantify the ability of the model to correctly identify the relevant aspects of customer needs. This makes it comparable to future research and creates a baseline for word-level Needmining models. Second, we discuss our contribution to literature and discuss some general aspects of the model output. Finally, we elaborate on how the output of our model can be used to guide decision makers. This entails a presentation of the raw model output and ideas for further enriching the data with relevant information, such as frequency and sentiment, to highlight important needs. Finally, we discuss some real-life efficiency implications based on Timoshenko and Hauser's (2019) [5] empirical evaluation of binary

Needmining.

5.1. Classifier performance

We evaluate the classifier based on k-fold cross-validation ($k = 5$), which is the process of randomizing all sampled data and then splitting it into k equally sized subsets. Then, we train k models, each based on a training set composed of $k-1$ subsets and using the remaining subset as the test set. This avoids an overly optimistic performance output caused by random chance during the train/test splitting of the data [45]. The final score is the average performance metrics between all k models.

The measurements we use are typical metrics that result from a classical confusion matrix, i.e., we differentiate each classification within the limits of being true positive (TP), true negative (TN), false positive (FP), and false negative (FN). Table 2 presents the mathematical representation of the commonly used metrics recall, precision, and the F1-score [46,47]. Precision is the ratio between the correctly identified positive (TP) samples, and all positive cases within the sample (TP + FP). Recall measures how many positive cases within the sample (TP + FN) the model was able to correctly identify (TP) [47]. Finally, the F1-score is the harmonic mean of precision and recall [48].

Needmining aims at extracting consumer needs from large bodies of user-generated content to improve product developer's efficiency when dealing with user-generated content. In a training and testing environment, we are interested in correctly identifying as many positive cases as possible. However, if only a small portion of identified cases reflect actual needs as identified by expert taggers, its impact on efficiency reduces because developers would have to review many cases that are not actual needs. This effect is measured by precision. Similarly, the artifact should find as many needs identified by expert taggers as possible, to reflect the information provided by expert taggers. If recall is low, the model misses a lot of needs, which experts previously identified as relevant. Given the F1-score is the harmonic mean between the precision and recall, it combines the two metrics into a single measurement reflecting two important factors for the overall success of the model from a product developer's point of view. Because there is no inherently easy way to grasp the meaning of the F1-score, we always include the individual scores for precision and recall. All reported metrics lie on a scale between 0 and 1, with scores closer to 1 being better.

The goal of this study is not to create a bottom-up approach for a new generation of language models, but instead to use an existing framework to show how token classification can provide new insights and possibilities for Needmining on the word level. Therefore, we train a model using the popular and well-known spacy framework for Named Entity Recognition, thus taking an already implemented, open source, "out-of-the-box" infrastructure to provide a proof of concept. To keep the training process as simple as possible, we adapt the optimizer equal to the suggestions provided by the spacy framework. This means starting with a learning rate of 0.00005 with 20,000 warm up steps. The optimizer is an implementation of Adam.v1, with a beta of 0.9, a beta2 of 0.999, and L2 weight decay of 0.01 starting at 0.01. As our primary focus is optimizing the F1-score, the weights for metric importance are set as 1 for F1 and 0 for both accuracy and recall.

It is possible to train this model type either on the review level, i.e., the training data are the fully annotated product review, or on the sentence level, where we first split each review into its individual

sentences. We tested both approaches and found that training on sentences leads to better results. Therefore, we report the performance for models trained on single sentences of the reviews. During the training process, we employ an early stopping algorithm based on the F1-score of the summary model. During training, the training process calculates this metric every two hundred training steps. As opposed to training each model for a fixed period, which would be more resource- and time-intensive, early stopping automatically quits the training process once the model does not improve over a certain number of iterations. We set this so-called early stopping patience at 1600 steps, i.e., the model will stop training once it cycles through 1600 steps without improving the F1-score. On average, the folds train for 183 epochs.

Table 3 displays the average performance of all k-folds for all entities and the 'attribute' and 'characteristic' token categories, based on different embeddings generated by pre-trained transformer models. RoBERTa performed the best with an F1-score of 71.2% across all entities, an average F1-score of 79.22% for attributes, and an average F1-score of 67.28% for characteristics. While BERT has a slight advantage over RoBERTa in the characteristics category, we found that most transformer-based embeddings perform similarly.

Additionally, we tested the token classifier's generalization capability by scraping and annotating 500 sentences in the "baby," "pets," and "sports" product categories. We tested those as they are substantially different from the training data, which were concerned with AirPod reviews. Table 4 summarizes the results of our generalization test. It shows three different metrics, each relating to a different level. The rationale behind using these levels is the fact that the token classifier is a method that returns true positives for *exact matches* between the annotation and model prediction. However, the model often identifies individual needs instead of exact word ranges specified by the annotations, resulting in false positives. Therefore, we introduced a new performance metric that measures token overlap. We further introduced the metric sentence match to compare our model to existing, binary Needmining models. Looking at the results, we can see that the differences among the levels are quite considerable. For the sports category, we obtained an F1-score of 49.11% for the exact match level, while achieving an F1-score of 77.03% on the sentence match level. Similar patterns can be observed for the other categories. Overall, these numbers indicate that our model generalizes well and outperforms existing, binary approaches.

5.2. Discussion

We reconceptualized the Needmining approach from a binary classification problem to a token classification problem and successfully created an artifact that extracts product attributes and characteristics from UGC. Table 4 shows how this artifact contributes to the Needmining literature by successfully providing a solution to the problem of identifying specific customer needs in UGC. Our token classification model identifies specific tokens within UGC that are representative of a need. Thus, we can extract the specific needs as an attribute-characteristic combination. Moreover, by removing all content without any identified attributes and characteristics, the artifact is also able to reduce the overall body of UGC to its informative content, effectively competing with existing Needmining methods.

In the following paragraphs, we take a detailed look at the output of our artifact and discuss its performance, the question of how to present the information produced by our artifact, and the economic impact and efficiency gains of our artifact.

5.3. Need identification

The best model achieves a cross-validated F1-score of 72.06% making its performance comparable to binary Needmining models while solving an inherently more complex task. It is able to correctly label a significant portion of both attributes and characteristics. However, it performs better at identifying attributes, which are typically shorter,

Table 2
Relevant performance metrics.

Precision	$\frac{TP}{TP + FP}$	Percentage of all positive values that is correctly identified as positive
Recall	$\frac{TP}{TP + FN}$	Percentage of all needs identified by experts uncovered by the model
F1-score	$2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$	Harmonic mean between precision and recall

Table 3

Token classifier cross validated performance by embedding.

		All entities			Attribute			Characteristic		
Embedding		Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
	RoBERTa	72.2%	70.2%	71.2%	79.7%	77.7%	78.7%	67.2%	65.3%	66.2%
	BERT	72%	69.8%	70.8%	79.3%	76%	77.5%	67.2%	65.7%	66.4%
	XLNet	71%	70.4%	70.7%	77.4%	78.2%	77.8%	66.6%	65.3%	65.9%
	ALBERT	69.8%	69.8%	69.8%	77.9%	76.2%	77%	64.5%	65.6%	65%
	DistilBERT	71.7%	67.5%	69.5%	79%	74.2%	76.4%	66.9%	63.1%	64.9%

Table 4

Token classifier performance in percent based on performance level.

		Baby			Pets			Sports		
Level		Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
	Sentence Match	69.75	92.62	79.58	68.53	71.01	69.75	67.7	89.34	77.03
	Token Overlap	66.67	79.14	72.37	62.41	54.19	58.01	64.11	74.11	68.75
	Exact Match	42.6	52.36	46.98	34.86	31.13	32.89	44.84	54.29	49.11

with an average length of 1.5 words compared to two words for characteristics. Characteristics can be more complex, potentially consisting of multiple tokens that may not semantically link together, such as the example “stay on the pizza”. This complexity increases the difficulty of identifying tokens correctly, as only completely correct spans count towards the true positive measure. Missing even part of a token negatively impacts all performance metrics, resulting in false positives.

The challenge is particularly pronounced when utilizing the AirPods-trained model for predicting needs in our generalization tasks. The exact match capabilities yield an average F1-score of only 42.99% across the categories ‘baby’, ‘pets’, and ‘sports’. However, the overlap performance, which identifies slightly different spans that are still informative for product developers, remains competitive with an average F1-score of 66.37% across categories. Fine-tuning should be able to easily elevate this metric above a 70% threshold. Finally, compared to binary Needmining, our model’s generalizability achieves a performance of 75.45%, making it comparable to models that specialize in binary Needmining [22,49].

The output of the token classifier is a list of all sentences including the identified spans and their predicted classes. In our example, there are 303 uniquely worded attributes in the AirPods data. However, the number of uniquely worded attributes is not representative of the actual number of attributes. The free-text nature of UGC allows customers to use different wordings to express the same information. For example, one frequently mentioned attribute is “noise cancelling” but the classifier also correctly identifies wordings such as “noise cancelation”, and “noise cancelation features” in addition to alternative spellings such as ‘cancelling out noise’, and ‘noise cancelling’. This is a common theme when analyzing UGC. To further process the attributes and

characteristics identified of our artifact, product developers can use clustering algorithms to group semantically related tokens. This would significantly reduce the number of attributes to be analyzed and may also highlight needs related to less frequently mentioned attributes. Applying pre-processing steps, such as stop word removal, stemming, and lemmatization [50], may even further improve the quality of the resulting clusters.

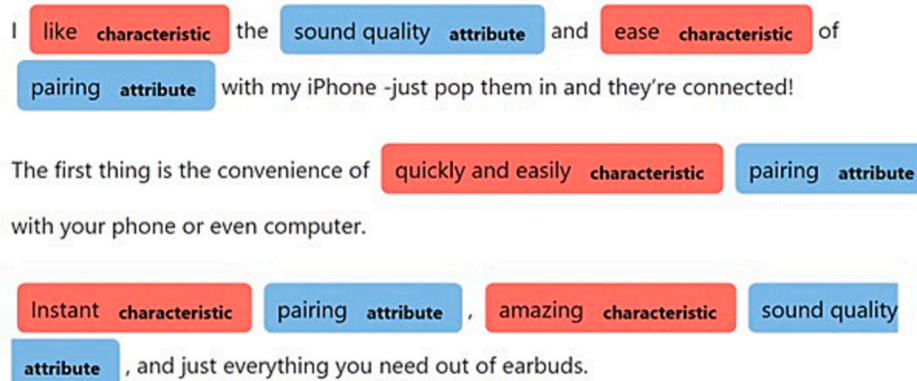
5.4. Information organization

Once the final results from our artifact are obtained, product developers need to decide how to organize the available information. While there is no inherently correct way of presenting the data, there are different approaches that can aid decision makers in tackling Needmining-related problems.

In Fig. 5, we present a way of organizing the information using color-code reviews. Presenting only sentences where the model detected a need fulfills the role of a binary classification model, while color-coding enhances the readability of reviews, which contributes to the efficiency of extracting needs manually.

Another option provided by our artifact is to produce a description of a product based on attributes and the attribute’s associated characteristics. This provides product developers with an explicit list of relevant attributes, how frequently each of these attributes appear in the reviews, and the experience customers may have had with the individual attributes as expressed by the sum of characteristics.

Such an attribute summary could be further enhanced by the use of sentiment analysis [24,28]. Sentiment analysis is a useful tool to provide additional information for product developers when evaluating the

**Fig. 5.** Artifact Output of Color-Coded Reviews for the Attribute “Pairing”.

discovered needs. It gives them insights into how positively or negatively customers feel towards an attribute based on which they can further investigate potential needs for future development. For future development, positive sentiment towards an attribute suggests that future iterations of the product should continue to embrace the attribute-characteristic combinations that customers liked. On the contrary, negative sentiment towards an attribute indicates room for improvement, if not a total lack of an expected feature.

To employ sentiment analysis in our setting, we give every sentence mentioning an attribute a positive or a negative tag. Then, we collect the sum of positive and negative tags for every attribute to determine a valence score, which reflects the relative number of positive sentiments compared to negative sentiment. Fig. 6 shows the relative proportion of positive and negative mentions for the twenty-five most mentioned attributes throughout the reviews on a scale from -1 to 1 where -1 denotes an entirely negative sentiment with 100% of the reviews being negative and 1 denotes an entirely positive sentiment with all reviews being positive.

Besides review sentences that mention characteristics without an associated attribute ('the product'), the attribute with the least positive sentiment is 'case'. Upon a closer inspection of the negative reviews associated with the 'case' attribute, we were able to identify a customer need that relates to wireless charging support for the case. Within the characteristics associated with the case, we found the two characteristics 'isn't wireless charging compatible' and 'doesn't charge wirelessly'. Here it is important to realize that the second generation of AirPods simply did not have this feature. However, by applying our artifact to product reviews, we were able to identify this as a potentially relevant customer need. Recall that we scraped our data in November 2019, well before the new AirPods were released in October 2021, which in fact supported this feature.

5.5. Economic impact and efficiency gains

Identifying the real-life economical example of the proposed artifact is complex because it requires significant knowledge on product development processes. However, we were able to demonstrate that our artifact can provide substantial support for a task that would otherwise be conducted manually. Assuming that processing a review manually takes on average 25 s means that our artifact can save almost 7 h of manual labor per 1000 reviews. Realizing that popular products, as discussed earlier, often have several hundreds of thousands of reviews, we believe that our artifact can significantly improve productivity. While our Needmining artifact needs to be trained based on annotated data, this only has to be done once. After training, our artifact can

predict specific customer needs for other sets of user generated content over and over, simply by changing the sample that is fed into the model. Additionally, it quantifies the occurrences of the needs, therefore offering decision support for product managers which are most relevant for the customer. With little additional effort, our artifact can also extract additional information, such as the average sentiment of the reviews. This can provide an overview of how customers think about certain aspects of the product, which is extremely useful to identify currently unmet needs by the market which offer innovation potential.

The key difference to other Needmining methods is that we extract the need-components directly instead of identifying the presence of a need within a sentence or review. Compared to other approaches, word-level Needmining may only require the full review of the content on a case-by-case basis. In binary Needmining, every identified sentence must still be manually reviewed. Word level Needmining reduces this effort, as it automatically quantifies the needs and allows experts to easily access sentences based on the needs that were identified by our model.

6. Limitations and future research

Naturally, our study is subject to some limitations. First, the current approach does not summarize similar or closely related attributes. This results in two of the top three attributes identified by frequency are 'sound' and 'sound quality'. In the future, we plan to summarize these by clustering their embeddings and combining all associated characteristics into attributes. This step allows product developers to save more time by reducing redundant information.

Second, the artifact only identifies all attributes and characteristics as they appear in a sentence. If a sentence contains multiple attributes, the current solution is to attach all characteristics to all attributes in the sentence, thus not exactly matching characteristics to the right attributes. While this approach does still provide an overview of relevant attribute-characteristic matches, simply based on the overall quantity a characteristic appears in the context of a given attribute, in the future we plan to refine the accuracy of attribute-characteristics matches using part-of-speech tagging within a sentence.

Finally, it should be noted that not all reviews are plain text. Many review platforms, including Amazon's product review section, give customers the opportunity to attach certain meta-data like star ratings or helpfulness ratings. We believe that causal inference from identified attributes and attribute-characteristic pairs to meta-data could provide useful information for product developers. For example, knowing that certain attributes for a product consistently reduces the star-rating of reviews by a given amount could suggest that product developers should

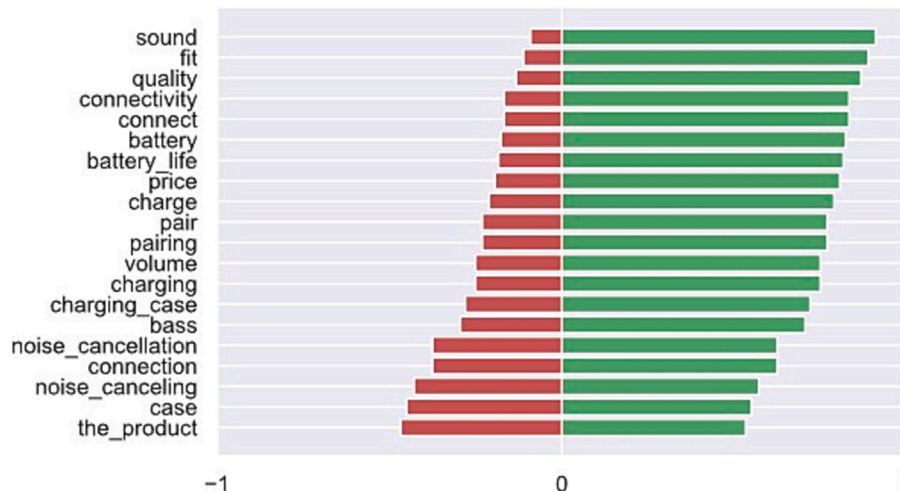


Fig. 6. Sentiment of 25 Most Frequently Mentioned Attributes.

focus on this attribute in future products to increase the overall performance of a product, especially given how important product reviews are for customer's purchase decisions.

7. Conclusion

This study is the first to reconceptualize Needmining from a binary classification to a token classification task. We disentangle the need concept into two components that are representative of a need. Attributes are the “what” in the product that customers discuss, and characteristics are the “how” customers talk about the attribute. For example, the sentence “The crust was too doughy” contains a need that can be represented by the attribute “crust” and the characteristic “too doughy”. We built an artifact that accurately identifies relevant tokens within UGC and classifies them according to the determined need components. In extension, the same approach can also binarily classify UGC with respect to need-content (Yes, No) by considering sentences with identified tokens as informative and those without as uninformative, thus effectively competing with previous Needmining models.

This contributes to the knowledge base in three ways. First, it provides an approach to solve the research gap identified in previous studies that the precise identification of customer needs requires additional automation. Second, the artifact itself is a novel and innovative use of token classification to address a previously unsolved problem. Third, we split the abstract concepts of a need into more explicit sub parts of attributes and characteristics. So far, current studies mostly use token classification to classify straight-forward concepts such as part-of-speech tags and entities (location, person, etc.). Biomedical research found novel uses of the concept in their own domain by identifying entities specific to their literature. We show additional usages of this method where one can decompose abstract concepts such as needs and then use token classification to identify the resulting components in unstructured textual data.

CRedit authorship contribution statement

Oliver Ettrich: Conceptualization, Methodology, Writing – original draft, Software, Validation, Formal analysis, Investigation, Writing – review & editing, Visualization. **Sven Stahlmann:** Conceptualization, Methodology, Software, Resources, Investigation, Data curation, Writing – review & editing, Visualization. **Henrik Leopold:** Writing – review & editing. **Christian Barrot:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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